

CMBSG Hardware Implementation and Development Process Guide

VOLUME 4

Commissioning, Startup & Performance Testing Protocols

REFERENCE DESIGN BASIS: 10,000 NM³/HR CMBSG MODULE
DOCUMENT CLASS: ENGINEERING DELIVERABLE — COMMISSIONING

Read Before Use

This is the volume where several open engineering questions carried forward from Volumes 1–3 get resolved with real data instead of assumption. **Section 7 (Stage 3 Capture-Efficiency Pilot Verification) is the single most important test in this entire commissioning program** — it directly tests whether the CA-KR1 enzyme delivers the mass-transfer enhancement the whole capture-efficiency claim depends on. **Do not proceed to full-scale continuous operation, and do not release the titanium mesh purchase order for any additional tower sections, until Section 7's test is complete and its decision gate is satisfied.** This volume assumes Volume 3 E&I completion has been certified for the stage(s) under commissioning.

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1 How to Use This Volume

1.1 Commissioning Philosophy

Commissioning proceeds in four escalating phases, each a gate for the next: **utility commissioning** (Section 2) → **dry commissioning** (no process fluids, Section 3) → **wet/cold commissioning** (water or inert surrogate fluids, Section 4) → **hot commissioning** (actual process chemistry, Sections 5–8). Do not skip a phase to save schedule — every phase exists because the failure modes it catches are cheaper to find before the next phase begins.

1.2 Relationship to Prior Volumes

This volume consumes the corrected process basis from Volume 1, the as-built mechanical configuration from Volume 2 (including the Stage 3 bolt-on extension provision, if fitted), and the instrumentation/control architecture from Volume 3 (including the additional high-resolution logging added specifically to support Section 7’s pilot test). It hands off to Volume 5 for final SIS proof-testing prior to continuous unsupervised operation, and to Volume 6 for the Standard Operating Procedures that govern day-to-day running once commissioning is complete.

2 Utility Commissioning

Table 1: Utility commissioning checklist.

Utility	Commissioning Check
Electrical power	Full-plant power-up per Volume 3 single-line diagram; verify all MCC sections energize, VFDs respond to a test speed command, UPS transfers cleanly on simulated grid loss
Instrument air	Verify dew point per ISO 8573-1 Class 3 or better, pressure stable at all pneumatic actuator locations (notably the Stage 1 bypass valve V-1, a safety-critical final element)
Cooling water	Verify flow and pressure at R-201 jacket, V-101 quench nozzle, and any other cooling-water consumer; check for cross-contamination risk at any shared header
Plant/instrument water	Verify quality (conductivity, pH) meets the Stage 1 quench makeup-water assumption used in Volume 1’s mass balance
Nitrogen (if used for HP-401 accumulator pre-charge)	Verify supply pressure and purity per vendor spec
Enzyme cold storage	Verify freezer stability at -20°C for minimum 48 hours unloaded before accepting CA-KR1 delivery (Volume 2, Section 2.3)

3 Dry Commissioning — No Process Fluids

3.1 Purpose

Dry commissioning exercises every motor, valve, and interlock without introducing gas, liquid, or chemical into the system. This is where wiring errors, incorrect valve fail-actions, and logic-configuration mistakes are cheapest to find.

3.2 Sequence

Dry Commissioning Sequence

1. **Loop checks:** for every tag in Volume 3's instrument tables, inject a simulated signal at the field device (or disconnect and use a calibrator) and verify correct display, engineering-unit scaling, and alarm activation at the HMI. Sign off each loop individually per Volume 3, Appendix B template.
2. **Motor rotation checks:** bump-test every motor (F-101, P-101, P-202, agitators, PM-401, HC-401 pump, HP-401 HPU) for correct rotation direction before coupling to driven equipment.
3. **Valve stroke checks:** stroke every control and isolation valve full open/full closed, verify position feedback matches commanded position, and time the stroke (important for the Stage 1 bypass valve V-1, which must open within the SIS-A required response time — confirm this with Volume 5 before final sign-off).
4. **Interlock/permissive checks:** with the process empty, simulate each SIS input (see Volume 3, Appendix C cause-and-effect skeleton) and verify the correct final element responds, **without** yet relying on this test as the formal SIL proof test (that is Volume 5's responsibility with its own documented procedure).
5. **HP-401 dry-cycle test:** run the press through a full compress/dwell/release/eject cycle with no material loaded, at reduced pressure first (per vendor commissioning procedure), then at full 200 bar, verifying cycle time and confirming no unexpected vibration or noise before any loaded cycle is attempted.

HOLD POINT — Dry Commissioning Sign-Off

Do not introduce any process fluid, water, or chemical into any stage until dry commissioning is 100% signed off for that stage. A loop-check failure found after water or chemical is in the system is significantly more expensive and, in Stage 2/3, may require full drain-down and re-cleaning before the fault can even be accessed.

4 Wet/Cold Commissioning — Water and Inert Surrogates

4.1 Purpose

Wet commissioning introduces water (and, where appropriate, an inert surrogate for the real process fluid) to verify hydraulic performance, level control, pump curves, and mechanical seal integrity before any hazardous or expensive chemical is introduced.

4.2 Stage 1 Wet Commissioning

Stage 1 Wet Commissioning Sequence

1. Fill T-101 sump with water to normal operating level; verify LIT-101 reading against a manual dip-check.
2. Run P-101 recirculation pump; verify flow rate against the pump curve and check for cavitation noise.
3. Introduce ambient-temperature air flow through D-101/V-101 (no combustion gas yet) at a reduced flow rate; verify PIT-102 tracks the expected ΔP curve. **Recall the corrected basis (Volume 1, Section 3.2): expect a much lower ΔP than the original reference's stated 1.47 kPa at full flow — do not force a false correlation to the uncorrected number.**
4. Ramp to full 10,000 Nm³/hr equivalent air flow (using ambient air, not yet hot combustion gas) and confirm F-101 VFD can achieve and hold the design flow rate.

4.3 Stage 2 Wet Commissioning

Stage 2 Wet Commissioning Sequence

1. Fill R-201 and T-202 with water; verify level instrumentation and agitator operation with no slag/chitosan charged yet.
2. Run P-202 recirculation loop with water; verify flow against Hagen-Poiseuille prediction (Volume 1: 0.115 bar over 25m DN80 CPVC at 30 m³/hr) and check for any unexpected pressure drop that could indicate a partially-closed valve or fouled static mixer element.
3. **Do not introduce acid, slag, or chitosan yet** — this is a water-only mechanical/hydraulic verification. Chemical introduction happens in Section 6 (Hot Commissioning).

4.4 Stage 3 Wet Commissioning

Stage 3 Wet Commissioning Sequence

1. Trickle water (no chitosan matrix, no enzyme) across all three mesh tiers at the design 2.5 L/s flow rate; visually verify even distribution across the full 2.5m×2.5m cross-section with no dry channeling or pooling.
2. Confirm ultrasonic transducers operate at the corrected 0.32 μm minimum amplitude target (Volume 1/2/3) with water present, checking for any unexpected cavitation erosion pattern on the mesh after a short trial run (inspect visually after draining).
3. Run ambient air through the tower (no CO₂-bearing gas yet) to verify PIT-301 tracks the expected pressure-drop curve at design flow.

4.5 Stage 4 Wet Commissioning

Stage 4 Wet Commissioning Sequence

1. Run PM-401 pug-mill with a water/inert-sand or water/limestone-powder surrogate mixture approximating the design solids loading, to verify shaft loading and discharge consistency without risking real process chemistry on a first run.
2. Run HC-401 hydrocyclone with the same surrogate; verify underflow solids concentration approaches the 65% target.
3. Run HP-401 with a surrogate-fed cavity (e.g., a standard concrete-block test mix, if available, or per vendor's own commissioning-test material recommendation) to verify press cycle and de-molding before committing real CMBSG gel material to the first loaded cycle.

5 Hot Commissioning — Introducing Process Chemistry

HOLD POINT — Hazardous Chemistry Introduction

Before introducing HF (mesh etching support chemistry, if any field touch-up is needed), NaBH₃CN, or live acid dosing, **Volume 5's HSE sign-off for the specific chemical handling procedure must be complete and the relevant personnel must have completed the training in Volume 8.** This volume sequences *when* hot commissioning happens; Volume 5 governs *how* it is done safely.

5.1 Stage 1 Hot Commissioning

Stage 1 Hot Commissioning Sequence

1. Introduce actual flue gas (or an equivalent test-fired gas source) at reduced temperature first, ramping toward the 180 °C design inlet condition in stages, monitoring TIT-101 and the ceramic liner (Volume 2, Section 3.4.2) for any thermal-shock indication.
2. Verify quench performance: outlet temperature approaches the 45 °C saturated design point (Volume 1, Section 3.2, verified-correct heat balance).
3. Verify PM removal qualitatively (visual stack observation or portable particulate monitor) — **this is the first opportunity to empirically check the PM10 removal claim flagged in Volume 1 (the stated 32 μm cut-diameter does not reproduce from the paper's own Calvert-model inputs). Record actual measured removal efficiency and compare against the 93–97% design target; do not assume the design target is met without this measurement.**

5.2 Stage 2 Hot Commissioning

Stage 2 Hot Commissioning Sequence

1. Charge R-201 with a reduced slag feed rate first, dose acid gradually, verify AIT-201 tracks toward the 2.5–3.5 pH target without overshoot.
2. Ramp to full 250 kg/hr slag feed rate; verify reactor temperature stays within the cooling jacket's capacity (mildly exothermic reaction per Volume 1).
3. Charge T-202 with chitosan solution, verify AIT-202 reaches the 4.0–4.5 pH target.
4. **Activate the AIC-202-TRIM loop (Volume 2/3 correction) and verify it responds correctly:** deliberately allow SI to drift (e.g., by briefly under-dosing acid) and confirm the trim loop engages before the main pH loop would have corrected it on its own. Record the response time and trim dosing volume for the commissioning report — **this is the practical verification of the saturation-index correction identified in Volume 1.**

6 Stage 3 Capture-Efficiency Pilot Verification

Why This Section Exists

Volume 1 found that the Hatta number computed from the original technical reference's own stated inputs is 0.088, not the claimed 0.877 (a $10\times$ decimal error), and that the resulting mass-transfer enhancement factor recomputes to approximately 1.0, not the claimed 1.25. If the corrected value is closer to reality, the CA-KR1 enzyme may not be delivering enough mass-transfer enhancement to reach the claimed 95–99% CO₂ capture efficiency at the stated 12 mg/L dosing and 27s residence time. **This section is a structured test protocol to settle that question with real plant data before any commercial or investment claim relies on the original (unverified) number.**

6.1 Test Matrix

Table 2: Stage 3 capture-efficiency test matrix.

Run	CA-KR1 (mg/L)	Residence Time	CO ₂ Feed (vol%)	Purpose
1	0 (no enzyme)	27s (12m tower)	14%	Baseline / uncatalyzed rate confirmation
2	5	27s	14%	Low-dose reference point (matches original Figure 3 curve)
3	12	27s	14%	Design point — direct test of the original claim
4	12	21.6s (reduced height, if extension not yet fitted)	14%	Sensitivity to residence time at design dosing
5	12	27s + extension (if fitted)	14%	Only run if Run 3 underperforms target
6	20	27s	14%	Upper-bound dosing sensitivity check

6.2 Test Procedure

Pilot Test Procedure — Each Run

1. Stabilize gas flow, temperature, and CO₂/SO₂ feed concentration at design conditions for minimum 30 minutes before recording data (allow the tower to reach steady state).
2. Set CA-KR1 dosing to the run's target concentration via DS-301; allow minimum 15 minutes for matrix composition to stabilize at the new dosing rate.
3. Log AIT-301 (inlet CO₂), AIT-302 (outlet CO₂), FIT-301-HR (matrix flow, high-resolution per Volume 3), CIT-301 (enzyme activity), TIT-301, and PIT-301 at the Volume 3-specified 5-second interval for a minimum 60-minute steady-state window.
4. Compute capture efficiency directly from AIT-301 / AIT-302 (not by back-calculating from an assumed kinetic model) as:

$$\eta_{\text{CO}_2} = \frac{[\text{CO}_2]_{\text{in}} - [\text{CO}_2]_{\text{out}}}{[\text{CO}_2]_{\text{in}}} \times 100\%$$

5. Close the mass balance independently using FIT-301-HR and the measured CaCO₃ production rate at Stage 4 (cross-check against Volume 1's corrected 5,934 kg/hr CaCO₃ basis at 95% capture) as a second, independent verification of the gas-phase measurement.
6. Repeat for each run in the test matrix; allow the system to re-stabilize between runs (minimum 30 minutes) before changing conditions.

6.3 Decision Gate

Table 3: Decision gate — action required based on Run 3 (design point) result.

Run 3 Result	Required Action
$\eta_{\text{CO}_2} \geq 93\%$	Original design claim substantially confirmed. Proceed to full performance test (Section 8) using the as-tested operating point.
$80\% \leq \eta_{\text{CO}_2} < 93\%$	Partial shortfall. Run Test 5 (tower extension, if fitted) or increase dosing per Run 6 result; re-test before committing to continuous operation at the original design point.
$\eta_{\text{CO}_2} < 80\%$	This would be consistent with the corrected Hatta-number finding (enhancement factor ≈ 1.0, i.e., little enzymatic benefit over the uncatalyzed baseline). Do not proceed to continuous operation at the original design point. Escalate to process engineering for a fundamental review: options include a substantially longer residence time (larger tower), a different enzyme loading strategy, or reassessing whether the enzyme's real-world k_{cat}/K_m match the vendor's stated values under actual matrix conditions (chitosan rheology may differ from the enzyme vendor's test conditions). Do not release any further Ti mesh purchase orders (Volume 1, Section 8) until this is resolved.

6.4 Cure-Time Sensitivity Trial (Stage 4, Run Concurrently)

Secondary Verification — Arrhenius Rate Constant

Volume 1 found the original reference's pozzolanic curing rate constant calculation contained an arithmetic error (stated 0.013 hr^{-1} ; correct value 0.082 hr^{-1} from the same inputs). A higher true rate constant suggests cure may complete faster than the original 12-hour assumption. Run three parallel curing-tunnel trials at 8, 10, and 12 hours residence time (adjust CT-401 conveyor speed accordingly), and test compressive strength at 48 hours for each batch (per Volume 1's verified-correct strength model). If 8 or 10 hours achieves the $\geq 25 \text{ MPa}$ M25 target, this shortens the production cycle and increases effective plant throughput at no additional equipment cost.

7 Integrated Full-Chain Commissioning

Integrated Commissioning Sequence

1. With Stage 3's design point confirmed per Section 7's decision gate, run all four stages simultaneously at reduced throughput (50% of design flow) for a minimum 8-hour continuous run.
2. Verify mass balance closure across the full chain: gas-side CO₂/SO₂ removal, Stage 4 solids production rate, and block output rate all trace consistently back to Volume 1's corrected basis.
3. Ramp to 75%, then 100% of design throughput, holding each level for a minimum 4-hour stabilization period before increasing further.
4. Verify all SIS functions remain armed and correctly configured throughout (coordinate with Volume 5 for the formal proof-test schedule, which may run concurrently with this integrated commissioning if resources allow).

8 Performance Test Protocol (Guarantee Test Run)

8.1 Test Conditions

Table 4: Performance test acceptance criteria, corrected basis.

Parameter	Acceptance Criterion	Basis
CO ₂ capture efficiency	As established by Section 7's decision gate — not the original reference's unverified 95–99% claim unless Run 3 confirmed it	Section 7
SO ₂ capture efficiency	≥95% (98% design target, this chain was verified correct in Volume 1)	Volume 1, Section 1.2
Block compressive strength (48hr)	≥25 MPa (M25 per IS-2185)	Volume 1, verified-correct strength model
Block production rate	≥1,900 blocks/hr (95% of the corrected 2,010 blocks/hr basis)	Volume 1, Section 1.4
Plant availability during test	≥95% uptime over the test window (excluding planned holds)	Standard commissioning practice

8.2 Test Duration and Reporting

Performance Test Execution

1. Minimum 72-hour continuous run at 100% design throughput, using the Stage 3 operating point established by Section 7.
2. Log all Section 2–3 utility parameters, Section 3–6 process parameters, and Section 7's capture-efficiency verification tags throughout.
3. Compile a performance test report comparing: (a) as-tested results, (b) the corrected Volume 1 design basis, and (c) the original technical reference's claims, with each discrepancy explicitly reconciled — **do not report only the favorable comparison**; if the plant underperforms the original reference's claims but meets the corrected Volume 1 basis, state both clearly.

9 Punch List Management

Table 5: Punch list category definitions.

Category	Definition	Handling
A	Safety-critical or prevents safe operation	Must close before hot commissioning of the affected stage
B	Affects performance or reliability but not immediate safety	Must close before Performance Test (Section 8) sign-off
C	Cosmetic, documentation, or minor	May be deferred to post-handover with engineering sign-off and a committed closure date

10 Commissioning Completion and Handover

Table 6: Commissioning completion checklist — gate for handover to Volume 6 (SOPs) and normal operation.

Item	Verification
All utility, dry, wet, and hot commissioning steps signed off	Section 2–6 records complete
Section 7 pilot verification decision gate satisfied	Test report on file, operating point confirmed
Performance test (Section 8) passed against corrected acceptance criteria	Test report on file
All Category A and B punch list items closed	Punch list document on file
Volume 5 SIS proof tests complete (coordinated, not owned by this volume)	Cross-reference to Volume 5 sign-off
Operator training (Volume 8) complete for initial operating crew	Training records on file

A Appendix A: Loop Check Sheet Template

Tag	Simulated Value	HMI Reading	Pass/Fail, Signature
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Table 7: Blank loop check sheet — one per instrument tag.

B Appendix B: Pilot Test Data Sheet Template

Run #	Timestamp	AIT-301	AIT-302	η_{CO_2}	Notes
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Table 8: Blank pilot test data sheet — log per Section 7 test procedure at 5-second intervals; this row format is for summary/steady-state averages.

C Appendix C: Cross-Reference to Prior Volume Corrections

Prior Finding	Commissioning Test Section (This Volume)
CO ₂ mass flow correction	Mass balance closure check during integrated commissioning uses corrected basis
Calvert d ₅₀ inconsistent	Empirical PM10 removal measurement during Stage 1 hot commissioning
Saturation index sign error	AIC-202-TRIM response verification test
Hatta number / enhancement factor unsupported	Full pilot verification test matrix and decision gate
Arrhenius rate constant error	Cure-time sensitivity trial (8/10/12hr)
Stage 1 ΔP correction	Wet commissioning ΔP curve check against corrected expectation